F/E/TZ

Reg. No:

Final Assessment Test – April 2026

VIT® Vellore Institute of Technology

Course: BCSE204L - Design and Analysis of Algorithms

Class NBR(s): 1876/1879/1881/1884/1886/1888/1890/ 1892/1895/1904/1907/1910/1912/1955/1967/1972/ 1984/1990/1997/2002/2008/2010/2015/2018/2640/ 2705/2734/4194

Slot: F1+TF1

Time: Three Hours

Max. Marks: 100

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

COs	CO Statements
CO1	Learn the fundamental concepts, principles, characteristics and performance measures of digital watermarking and steganography.
CO2	Acquire the various concepts of watermarking for digital authentication and authorization schemes related to electronic documents, image and video.
CO3	Gathering the various concepts of steganography to access the sensitive information concealing of message, image, audio or video within another file.
CO4	Design and implement efficient data hiding methods against steganalysis techniques.

BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Answer ALL Questions

(10 X 10 = 100 Marks)

1. a) Compute time complexity of the following recurrence relations using the Master Theorem. [4+6] CO1 BL2

   i. $T(n) = 4T(n/2) + n^2 \log n$

   ii. $T(n) = 2T(n/2) + n \log n$

   b) A thief can carry a maximum weight of 60 kg with the following items as shown below.

Item	1	2	3	4	5	6	7
Profit	60	100	120	80	75	90	30
Weight	10	20	30	40	15	25	5

Obtain the maximum amount that the thief can steal. Use a greedy algorithm to get the optimal solution.

Compare the greedy strategy for the 0–1 Knapsack vs the Fractional Knapsack. Prove why the value-per-weight greedy approach is optimal for the fractional version but fails for the 0–1 version. Support your answer with a detailed calculation for the above given scenario.

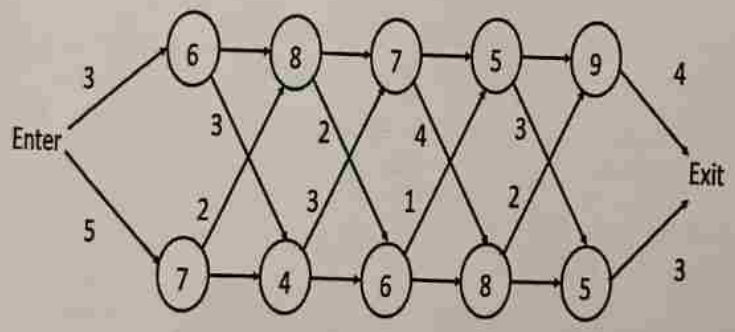
2. Multiply the two integers 1234 and 5678 using the Karatsuba's multiplication algorithm. Show all recursive calls up to the base case. While tracing the algorithm state the recurrence relation and compute the time complexity. **CO1 BL2**

3. A departmental store has 4 items. The following table shows the corresponding profit and weights of the items. **CO2 BL3**

Item	1	2	3	4
Weight	2	5	10	5
Profit	40	30	50	10

A person having a knapsack of size 10. Obtain a LIFO branch and bound technique to get the maximum profit. Construct the state space tree. Clearly indicate the live nodes, dead nodes. Show the stack (LIFO) operations at each step. Identify all feasible and infeasible solutions encountered. Determine the optimal solution.

4. Consider an assembly line system shown below, with two production lines, each having 5 stations. The processing times, transfer times, entry times, and exit times are stated in the figure. **CO2 BL3**



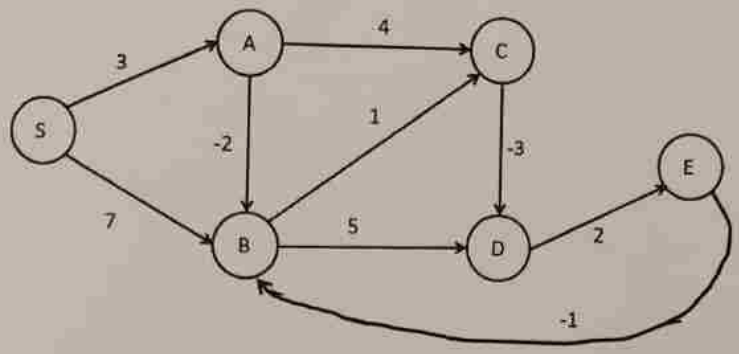
The processing times at each station are inscribed in the nodes. A product starts at the entry of either line and moves through all stations sequentially to reach the exit.

Obtain an optimal solution using dynamic programming. Illustrate the sequence of the steps clearly.

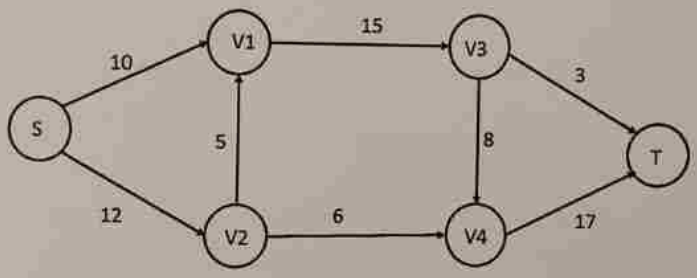
5. Identify the pattern P = 31415 in the text T = 2359023141526739921 using Robin-Krap algorithm. Consider base d=10 and prime modulus=11. **CO2 BL3**

i. Compute the hash value of the pattern.
ii. Compute rolling hash values for each substring of length equal to the pattern.
iii. Identify all positions where hash values match.
iv. Determine which matches are valid and which spurious hits.
v. Explain why spurious hits occur.

6. Find shortest distances from source S other vertices using Bellman-Ford Algorithm for the graph shown in the figure below. Do Bellman Ford algorithm is used to compute all pair shortest path. If yes, write the changes in the algorithm. **CO3 BL3**



7. Consider the flow network given below, where edges are labelled with maximum capacity. **CO3 BL4**



i. Apply the push–relabel algorithm to compute the maximum flow from source S to sink T.

ii. Show the initial preflow and the height of each node at each iteration.

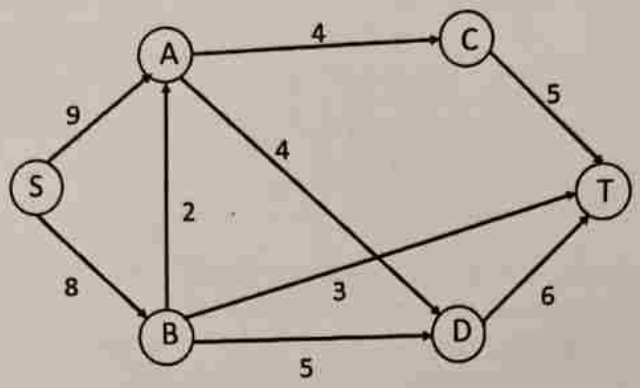
8. Given the following set of points in a 2D plane: **CO3 BL3**

P = {(1,1), (2,5), (3,3), (5,3), (3,2), (2,2), (4,4), (6,2), (7,4), (5,5), (6,6), (3,6), (1,4), (4,1)}. Identify the **pivot point** and apply the **Graham Scan Algorithm** to construct the convex hull. Show the status of the stack at every step. For each point, indicate whether a push or pop operation occurs. Support your answer with computations and diagram.

9.a) Apply randomized quick sort algorithm to sort A = [12, 4, 9, 1, 15, 7, 3, 10, 5, 8]. Illustrate the steps using the following pivots in order: 9, 4, 12, 7. A step-by-step tracing of the algorithm is expected. **CO4 BL3**

OR

9.b) Consider the following flow network shown in below. Verify the flow along the cut is equal to the maximum flow in the flow network. Consider the source vertex as S and sink vertex as T. **CO4 BL3**



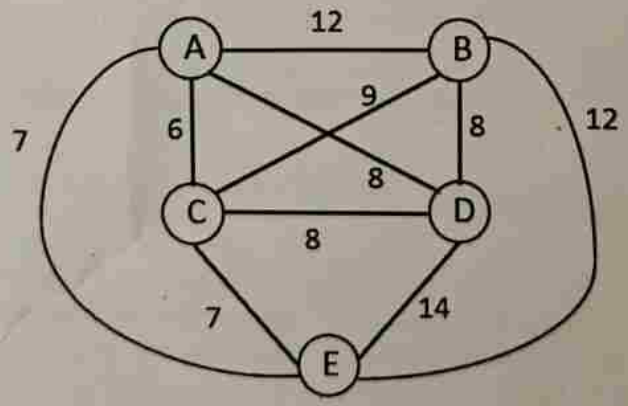
10.a) i. The independent set problem is defined as follows: **CO5 BL3**

Given an undirected graph G = (V, E) and an integer k, determine whether there exists a subset of vertices $V' \subseteq V$ such that $|V'| \geq k$ and no two vertices in V' are adjacent. Prove that the independent set problem belongs to NP -complete.

ii. Show correspondence between clique and independent set problem.

OR

10.b) Show that the travelling salesman problem is a polynomial time two approximation algorithm. Illustrate the above statement with the help of following graph. **CO5 BL3**



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